Daily:

MetroArea	TripCategory	DayOfWeek	TripCount	PercentWithinCategory
Austin	Commuter	Monday	34428	19.36
Austin	Commuter	Tuesday	37592	21.14
Austin	Commuter	Wednesday	36669	20.62
Austin	Commuter	Thursday	36730	20.66
Austin	Commuter	Friday	32395	18.22
Austin	Leisure	Monday	128525	11.83
Austin	Leisure	Tuesday	110994	10.22
Austin	Leisure	Wednesday	113307	10.43
Austin	Leisure	Thursday	125897	11.59
Austin	Leisure	Friday	163408	15.05
Austin	Leisure	Saturday	241507	22.24
Austin	Leisure	Sunday	202376	18.63
Los Angeles	Commuter	Monday	45627	19.01
Los Angeles	Commuter	Tuesday	51025	21.25
Los Angeles	Commuter	Wednesday	49879	20.78
Los Angeles	Commuter	Thursday	49434	20.59
Los Angeles	Commuter	Friday	44112	18.37
Los Angeles	Leisure	Monday	106299	12.33
Los Angeles	Leisure	Tuesday	110526	12.82
Los Angeles	Leisure	Wednesday	108677	12.61
Los Angeles	Leisure	Thursday	112731	13.08
Los Angeles	Leisure	Friday	124014	14.39
Los Angeles	Leisure	Saturday	151774	17.61
Los Angeles	Leisure	Sunday	147970	17.17
Pittsburgh	Commuter	Monday	9445	18.68
Pittsburgh	Commuter	Tuesday	10705	21.18
Pittsburgh	Commuter	Wednesday	10736	21.24
Pittsburgh	Commuter	Thursday	10391	20.56
Pittsburgh	Commuter	Friday	9274	18.35
Pittsburgh	Leisure	Monday	52910	12.16
Pittsburgh	Leisure	Tuesday	52103	11.98
Pittsburgh	Leisure	Wednesday	54895	12.62

Pittsburgh	Leisure	Thursday	51402	11.81
Pittsburgh	Leisure	Friday	57939	13.32
Pittsburgh	Leisure	Saturday	87176	20.04
Pittsburgh	Leisure	Sunday	78665	18.08
Portland	Commuter	Monday	29634	19.17
Portland	Commuter	Tuesday	31780	20.55
Portland	Commuter	Wednesday	32265	20.87
Portland	Commuter	Thursday	31944	20.66
Portland	Commuter	Friday	28991	18.75
Portland	Leisure	Monday	142368	12.6
Portland	Leisure	Tuesday	139326	12.33
Portland	Leisure	Wednesday	142590	12.62
Portland	Leisure	Thursday	146715	12.98
Portland	Leisure	Friday	166030	14.69
Portland	Leisure	Saturday	209343	18.52
Portland	Leisure	Sunday	183926	16.27

Monthly:

MetroArea	TripCategory	Year	Month	TripCount
Austin	Commuter	2013	12	21
Austin	Commuter	2014	1	274
Austin	Commuter	2014	2	388
Austin	Commuter	2014	3	975
Austin	Commuter	2014	4	1102
Austin	Commuter	2014	5	1094
Austin	Commuter	2014	6	1180
Austin	Commuter	2014	7	1371
Austin	Commuter	2014	8	1300
Austin	Commuter	2014	9	1412
Austin	Commuter	2014	10	1847
Austin	Commuter	2014	11	1048
Austin	Commuter	2014	12	958

Austin	Commuter	2015	1	1036
Austin	Commuter	2015	2	1025
Austin	Commuter	2015	3	1786
Austin	Commuter	2015	4	1684
Austin	Commuter	2015	5	1307
Austin	Commuter	2015	6	1661
Austin	Commuter	2015	7	2068
Austin	Commuter	2015	8	2023
Austin	Commuter	2015	9	2334
Austin	Commuter	2015	10	2591
Austin	Commuter	2015	11	1992
Austin	Commuter	2015	12	1759
Austin	Commuter	2016	1	3533
Austin	Commuter	2016	2	1908
Austin	Commuter	2016	3	2704
Austin	Commuter	2016	5	2026
Austin	Commuter	2016	6	2192
Austin	Commuter	2016	7	1994
Austin	Commuter	2016	8	2025
Austin	Commuter	2016	9	2251
Austin	Commuter	2016	10	2339
Austin	Commuter	2016	11	2003
Austin	Commuter	2017	1	1792
Austin	Commuter	2017	2	2085
Austin	Commuter	2017	3	3395
Austin	Commuter	2017	4	2348
Austin	Commuter	2017	5	2677
Austin	Commuter	2017	6	2748
Austin	Commuter	2017	7	2188
Austin	Commuter	2017	8	2227
Austin	Commuter	2017	9	2175
Austin	Commuter	2017	10	2562
Austin	Commuter	2017	11	2279

Austin	Commuter	2017	12	1364
Austin	Commuter	2018	1	1654
Austin	Commuter	2018	2	5008
Austin	Commuter	2018	3	11339
Austin	Commuter	2018	4	11049
Austin	Commuter	2018	5	7265
Austin	Commuter	2018	6	5063
Austin	Commuter	2018	7	4441
Austin	Commuter	2018	8	4747
Austin	Commuter	2018	9	7252
Austin	Commuter	2018	10	4957
Austin	Commuter	2018	11	3995
Austin	Commuter	2018	12	1927
Austin	Commuter	2019	1	2095
Austin	Commuter	2019	2	2237
Austin	Commuter	2019	3	2386
Austin	Commuter	2019	4	2380
Austin	Commuter	2019	5	1721
Austin	Commuter	2019	6	1298
Austin	Commuter	2019	7	1316
Austin	Commuter	2019	8	1234
Austin	Commuter	2019	9	1672
Austin	Commuter	2019	10	1490
Austin	Commuter	2019	11	851
Austin	Commuter	2019	12	607
Austin	Commuter	2020	1	748
Austin	Commuter	2020	2	843
Austin	Commuter	2020	3	894
Austin	Commuter	2020	4	580
Austin	Commuter	2020	5	664
Austin	Commuter	2020	6	1050
Austin	Leisure	2013	12	1436
Austin	Leisure	2014	1	3100

Austin	Leisure	2014	2	5118
Austin	Leisure	2014	3	24813
Austin	Leisure	2014	4	13355
Austin	Leisure	2014	5	13952
Austin	Leisure	2014	6	12470
Austin	Leisure	2014	7	12973
Austin	Leisure	2014	8	9828
Austin	Leisure	2014	9	10151
Austin	Leisure	2014	10	20945
Austin	Leisure	2014	11	9718
Austin	Leisure	2014	12	6315
Austin	Leisure	2015	1	6728
Austin	Leisure	2015	2	8541
Austin	Leisure	2015	3	28324
Austin	Leisure	2015	4	14870
Austin	Leisure	2015	5	14531
Austin	Leisure	2015	6	12232
Austin	Leisure	2015	7	14260
Austin	Leisure	2015	8	12500
Austin	Leisure	2015	9	12909
Austin	Leisure	2015	10	20269
Austin	Leisure	2015	11	9764
Austin	Leisure	2015	12	8294
Austin	Leisure	2016	1	18627
Austin	Leisure	2016	2	11447
Austin	Leisure	2016	3	26861
Austin	Leisure	2016	5	17531
Austin	Leisure	2016	6	14345
Austin	Leisure	2016	7	14261
Austin	Leisure	2016	8	10359
Austin	Leisure	2016	9	13684
Austin	Leisure	2016	10	21926
Austin	Leisure	2016	11	12650

Austin	Leisure	2017	1	8664
Austin	Leisure	2017	2	11514
Austin	Leisure	2017	3	23137
Austin	Leisure	2017	4	20056
Austin	Leisure	2017	5	16443
Austin	Leisure	2017	6	13358
Austin	Leisure	2017	7	12454
Austin	Leisure	2017	8	8915
Austin	Leisure	2017	9	12846
Austin	Leisure	2017	10	17890
Austin	Leisure	2017	11	12820
Austin	Leisure	2017	12	7537
Austin	Leisure	2018	1	10257
Austin	Leisure	2018	2	26454
Austin	Leisure	2018	3	63333
Austin	Leisure	2018	4	53584
Austin	Leisure	2018	5	35134
Austin	Leisure	2018	6	19714
Austin	Leisure	2018	7	17094
Austin	Leisure	2018	8	18096
Austin	Leisure	2018	9	30039
Austin	Leisure	2018	10	21551
Austin	Leisure	2018	11	16505
Austin	Leisure	2018	12	9977
Austin	Leisure	2019	1	7505
Austin	Leisure	2019	2	8874
Austin	Leisure	2019	3	13540
Austin	Leisure	2019	4	11040
Austin	Leisure	2019	5	9459
Austin	Leisure	2019	6	6232
Austin	Leisure	2019	7	7002
Austin	Leisure	2019	8	4855
Austin	Leisure	2019	9	6920

Austin	Leisure	2019	10	7343
Austin	Leisure	2019	11	4919
Austin	Leisure	2019	12	4336
Austin	Leisure	2020	1	3962
Austin	Leisure	2020	2	5207
Austin	Leisure	2020	3	6153
Austin	Leisure	2020	4	4460
Austin	Leisure	2020	5	8915
Austin	Leisure	2020	6	8833
Los Angeles	Commuter	2016	7	2840
Los Angeles	Commuter	2016	8	4979
Los Angeles	Commuter	2016	9	4443
Los Angeles	Commuter	2016	10	4190
Los Angeles	Commuter	2016	11	3750
Los Angeles	Commuter	2016	12	2711
Los Angeles	Commuter	2017	1	2407
Los Angeles	Commuter	2017	2	2525
Los Angeles	Commuter	2017	3	3612
Los Angeles	Commuter	2017	4	3276
Los Angeles	Commuter	2017	5	5199
Los Angeles	Commuter	2017	6	5146
Los Angeles	Commuter	2017	7	5037
Los Angeles	Commuter	2017	8	6089
Los Angeles	Commuter	2017	9	5488
Los Angeles	Commuter	2017	10	6121
Los Angeles	Commuter	2017	11	4843
Los Angeles	Commuter	2017	12	3777
Los Angeles	Commuter	2018	1	4660
Los Angeles	Commuter	2018	2	4539
Los Angeles	Commuter	2018	3	4708
Los Angeles	Commuter	2018	4	5030
Los Angeles	Commuter	2018	5	5984
Los Angeles	Commuter	2018	6	5838

Los Angeles	Commuter	2018	7	5326
Los Angeles	Commuter	2018	8	5788
Los Angeles	Commuter	2018	9	5095
Los Angeles	Commuter	2018	10	6290
Los Angeles	Commuter	2018	11	4934
Los Angeles	Commuter	2018	12	3699
Los Angeles	Commuter	2019	1	4539
Los Angeles	Commuter	2019	2	4202
Los Angeles	Commuter	2019	3	4244
Los Angeles	Commuter	2019	4	4651
Los Angeles	Commuter	2019	5	4624
Los Angeles	Commuter	2019	6	5554
Los Angeles	Commuter	2019	7	6284
Los Angeles	Commuter	2019	8	7232
Los Angeles	Commuter	2019	9	7013
Los Angeles	Commuter	2019	10	7635
Los Angeles	Commuter	2019	11	6042
Los Angeles	Commuter	2019	12	5443
Los Angeles	Commuter	2020	1	7486
Los Angeles	Commuter	2020	2	7180
Los Angeles	Commuter	2020	3	4648
Los Angeles	Commuter	2020	4	2845
Los Angeles	Commuter	2020	5	3266
Los Angeles	Commuter	2020	6	2385
Los Angeles	Commuter	2020	7	2383
Los Angeles	Commuter	2020	8	1969
Los Angeles	Commuter	2020	9	2128
Los Angeles	Leisure	2016	7	8580
Los Angeles	Leisure	2016	8	19174
Los Angeles	Leisure	2016	9	15423
Los Angeles	Leisure	2016	10	13969
Los Angeles	Leisure	2016	11	10677
Los Angeles	Leisure	2016	12	7901

Los Angeles	Leisure	2017	1	7940
Los Angeles	Leisure	2017	2	7006
Los Angeles	Leisure	2017	3	10290
Los Angeles	Leisure	2017	4	10583
Los Angeles	Leisure	2017	5	14181
Los Angeles	Leisure	2017	6	13533
Los Angeles	Leisure	2017	7	17049
Los Angeles	Leisure	2017	8	19118
Los Angeles	Leisure	2017	9	19556
Los Angeles	Leisure	2017	10	21703
Los Angeles	Leisure	2017	11	17789
Los Angeles	Leisure	2017	12	16981
Los Angeles	Leisure	2018	1	17561
Los Angeles	Leisure	2018	2	15448
Los Angeles	Leisure	2018	3	18471
Los Angeles	Leisure	2018	4	19833
Los Angeles	Leisure	2018	5	20131
Los Angeles	Leisure	2018	6	20541
Los Angeles	Leisure	2018	7	25024
Los Angeles	Leisure	2018	8	27540
Los Angeles	Leisure	2018	9	26510
Los Angeles	Leisure	2018	10	22635
Los Angeles	Leisure	2018	11	19019
Los Angeles	Leisure	2018	12	17290
Los Angeles	Leisure	2019	1	14932
Los Angeles	Leisure	2019	2	12763
Los Angeles	Leisure	2019	3	16666
Los Angeles	Leisure	2019	4	14077
Los Angeles	Leisure	2019	5	14281
Los Angeles	Leisure	2019	6	21321
Los Angeles	Leisure	2019	7	22204
Los Angeles	Leisure	2019	8	25597
Los Angeles	Leisure	2019	9	23746

Los Angeles	Leisure	2019	10	22534
Los Angeles	Leisure	2019	11	17989
Los Angeles	Leisure	2019	12	16653
Los Angeles	Leisure	2020	1	20710
Los Angeles	Leisure	2020	2	21607
Los Angeles	Leisure	2020	3	14744
Los Angeles	Leisure	2020	4	13437
Los Angeles	Leisure	2020	5	18240
Los Angeles	Leisure	2020	6	14013
Los Angeles	Leisure	2020	7	13995
Los Angeles	Leisure	2020	8	11668
Los Angeles	Leisure	2020	9	9358
Pittsburgh	Commuter	2017	1	336
Pittsburgh	Commuter	2017	2	467
Pittsburgh	Commuter	2017	3	468
Pittsburgh	Commuter	2017	4	762
Pittsburgh	Commuter	2017	5	921
Pittsburgh	Commuter	2017	6	1315
Pittsburgh	Commuter	2017	7	1160
Pittsburgh	Commuter	2017	8	1438
Pittsburgh	Commuter	2017	9	1299
Pittsburgh	Commuter	2017	10	1322
Pittsburgh	Commuter	2017	11	855
Pittsburgh	Commuter	2017	12	455
Pittsburgh	Commuter	2018	1	335
Pittsburgh	Commuter	2018	2	472
Pittsburgh	Commuter	2018	3	475
Pittsburgh	Commuter	2018	4	880
Pittsburgh	Commuter	2018	5	1728
Pittsburgh	Commuter	2018	6	1497
Pittsburgh	Commuter	2018	7	1727
Pittsburgh	Commuter	2018	8	1829
Pittsburgh	Commuter	2018	9	1490

Pittsburgh	Commuter	2018	10	1464
Pittsburgh	Commuter	2018	11	691
Pittsburgh	Commuter	2018	12	562
Pittsburgh	Commuter	2019	1	594
Pittsburgh	Commuter	2019	2	555
Pittsburgh	Commuter	2019	3	874
Pittsburgh	Commuter	2019	4	1500
Pittsburgh	Commuter	2019	5	2190
Pittsburgh	Commuter	2019	6	1992
Pittsburgh	Commuter	2019	7	2369
Pittsburgh	Commuter	2019	8	2516
Pittsburgh	Commuter	2019	9	2759
Pittsburgh	Commuter	2019	10	2453
Pittsburgh	Commuter	2019	11	1237
Pittsburgh	Commuter	2019	12	718
Pittsburgh	Commuter	2020	1	1042
Pittsburgh	Commuter	2020	2	890
Pittsburgh	Commuter	2020	3	646
Pittsburgh	Commuter	2020	4	254
Pittsburgh	Commuter	2020	5	267
Pittsburgh	Commuter	2020	6	551
Pittsburgh	Commuter	2020	7	778
Pittsburgh	Commuter	2020	8	1193
Pittsburgh	Commuter	2020	9	1225
Pittsburgh	Leisure	2015	5	480
Pittsburgh	Leisure	2015	6	8742
Pittsburgh	Leisure	2015	7	13886
Pittsburgh	Leisure	2015	8	13764
Pittsburgh	Leisure	2015	9	10459
Pittsburgh	Leisure	2015	10	7244
Pittsburgh	Leisure	2015	11	4651
Pittsburgh	Leisure	2015	12	3239
Pittsburgh	Leisure	2016	1	1645

Pittsburgh	Leisure	2016	2	2765
Pittsburgh	Leisure	2016	3	5433
Pittsburgh	Leisure	2016	4	6839
Pittsburgh	Leisure	2016	5	8857
Pittsburgh	Leisure	2016	6	10677
Pittsburgh	Leisure	2016	7	10910
Pittsburgh	Leisure	2016	8	8772
Pittsburgh	Leisure	2016	9	9196
Pittsburgh	Leisure	2016	10	6191
Pittsburgh	Leisure	2016	11	3811
Pittsburgh	Leisure	2016	12	1323
Pittsburgh	Leisure	2017	1	1215
Pittsburgh	Leisure	2017	2	2557
Pittsburgh	Leisure	2017	3	2181
Pittsburgh	Leisure	2017	4	6086
Pittsburgh	Leisure	2017	5	6878
Pittsburgh	Leisure	2017	6	8461
Pittsburgh	Leisure	2017	7	8922
Pittsburgh	Leisure	2017	8	7630
Pittsburgh	Leisure	2017	9	7257
Pittsburgh	Leisure	2017	10	5529
Pittsburgh	Leisure	2017	11	2228
Pittsburgh	Leisure	2017	12	1258
Pittsburgh	Leisure	2018	1	697
Pittsburgh	Leisure	2018	2	1278
Pittsburgh	Leisure	2018	3	1929
Pittsburgh	Leisure	2018	4	4191
Pittsburgh	Leisure	2018	5	9259
Pittsburgh	Leisure	2018	6	9418
Pittsburgh	Leisure	2018	7	11435
Pittsburgh	Leisure	2018	8	9774
Pittsburgh	Leisure	2018	9	7010
Pittsburgh	Leisure	2018	10	4294

Pittsburgh	Leisure	2018	11	1962
Pittsburgh	Leisure	2018	12	1475
Pittsburgh	Leisure	2019	1	1722
Pittsburgh	Leisure	2019	2	2063
Pittsburgh	Leisure	2019	3	4051
Pittsburgh	Leisure	2019	4	8115
Pittsburgh	Leisure	2019	5	11939
Pittsburgh	Leisure	2019	6	13005
Pittsburgh	Leisure	2019	7	13016
Pittsburgh	Leisure	2019	8	12879
Pittsburgh	Leisure	2019	9	12353
Pittsburgh	Leisure	2019	10	8365
Pittsburgh	Leisure	2019	11	4363
Pittsburgh	Leisure	2019	12	2677
Pittsburgh	Leisure	2020	1	2705
Pittsburgh	Leisure	2020	2	2777
Pittsburgh	Leisure	2020	3	4143
Pittsburgh	Leisure	2020	4	4452
Pittsburgh	Leisure	2020	5	12228
Pittsburgh	Leisure	2020	6	12009
Pittsburgh	Leisure	2020	7	13047
Pittsburgh	Leisure	2020	8	15209
Pittsburgh	Leisure	2020	9	12164
Portland	Commuter	2016	7	2419
Portland	Commuter	2016	8	6240
Portland	Commuter	2016	9	5829
Portland	Commuter	2016	10	3958
Portland	Commuter	2016	11	3715
Portland	Commuter	2016	12	2019
Portland	Commuter	2017	1	1956
Portland	Commuter	2017	2	2206
Portland	Commuter	2017	3	2935
Portland	Commuter	2017	4	3576

Portland	Commuter	2017	5	6087
Portland	Commuter	2017	6	6814
Portland	Commuter	2017	7	7083
Portland	Commuter	2017	8	7280
Portland	Commuter	2017	9	5284
Portland	Commuter	2017	10	4843
Portland	Commuter	2017	11	3598
Portland	Commuter	2017	12	2682
Portland	Commuter	2018	1	2978
Portland	Commuter	2018	2	2977
Portland	Commuter	2018	3	3984
Portland	Commuter	2018	4	4403
Portland	Commuter	2018	5	6224
Portland	Commuter	2018	6	4539
Portland	Commuter	2018	7	3931
Portland	Commuter	2018	8	3805
Portland	Commuter	2018	9	3120
Portland	Commuter	2018	10	3566
Portland	Commuter	2018	11	2368
Portland	Commuter	2018	12	1610
Portland	Commuter	2019	1	2031
Portland	Commuter	2019	2	1549
Portland	Commuter	2019	3	2253
Portland	Commuter	2019	4	2658
Portland	Commuter	2019	5	3179
Portland	Commuter	2019	6	2438
Portland	Commuter	2019	7	2745
Portland	Commuter	2019	8	2741
Portland	Commuter	2019	9	2114
Portland	Commuter	2019	10	2051
Portland	Commuter	2019	11	1506
Portland	Commuter	2019	12	1251
Portland	Commuter	2020	1	1282

Portland	Commuter	2020	2	1406
Portland	Commuter	2020	3	902
Portland	Commuter	2020	4	374
Portland	Commuter	2020	5	419
Portland	Commuter	2020	6	532
Portland	Commuter	2020	7	607
Portland	Commuter	2020	8	547
Portland	Leisure	2016	7	23589
Portland	Leisure	2016	8	45351
Portland	Leisure	2016	9	33192
Portland	Leisure	2016	10	15176
Portland	Leisure	2016	11	12201
Portland	Leisure	2016	12	5786
Portland	Leisure	2017	1	4590
Portland	Leisure	2017	2	6389
Portland	Leisure	2017	3	10455
Portland	Leisure	2017	4	16691
Portland	Leisure	2017	5	31167
Portland	Leisure	2017	6	33341
Portland	Leisure	2017	7	47328
Portland	Leisure	2017	8	39769
Portland	Leisure	2017	9	27924
Portland	Leisure	2017	10	19986
Portland	Leisure	2017	11	10473
Portland	Leisure	2017	12	8330
Portland	Leisure	2018	1	9318
Portland	Leisure	2018	2	9628
Portland	Leisure	2018	3	17890
Portland	Leisure	2018	4	17818
Portland	Leisure	2018	5	72185
Portland	Leisure	2018	6	43756
Portland	Leisure	2018	7	52592
Portland	Leisure	2018	8	42094

Portland	Leisure	2018	9	32163
Portland	Leisure	2018	10	23917
Portland	Leisure	2018	11	17753
Portland	Leisure	2018	12	13244
Portland	Leisure	2019	1	14650
Portland	Leisure	2019	2	10401
Portland	Leisure	2019	3	24063
Portland	Leisure	2019	4	25579
Portland	Leisure	2019	5	37712
Portland	Leisure	2019	6	36460
Portland	Leisure	2019	7	37919
Portland	Leisure	2019	8	36760
Portland	Leisure	2019	9	24604
Portland	Leisure	2019	10	21219
Portland	Leisure	2019	11	16173
Portland	Leisure	2019	12	11413
Portland	Leisure	2020	1	10949
Portland	Leisure	2020	2	13371
Portland	Leisure	2020	3	9377
Portland	Leisure	2020	4	7385
Portland	Leisure	2020	5	10787
Portland	Leisure	2020	6	11672
Portland	Leisure	2020	7	13504
Portland	Leisure	2020	8	12204

MetroArea	StdDevMonthlyCommuterSharePct
Austin	4.214103204
Los Angeles	3.419881555
Pittsburgh	9.880639722
Portland	6.876680388

Rain Resilience:

MetroArea	TripCategory	PctTripsInRainHours	PctRainyHours	RainParticipationRatio
Austin	Commuter	4.66	6.48	0.719
Austin	Leisure	3.65	6.48	0.564
Los Angeles	Commuter	1.96	3.68	0.532
Los Angeles	Leisure	1.78	3.68	0.485
Pittsburgh	Commuter	9.8	10.14	0.967
Pittsburgh	Leisure	5.68	10.14	0.56
Portland	Commuter	11.78	16.87	0.698
Portland	Leisure	9.05	16.87	0.537

Rain Contingency Table:

MetroArea	TripCategory	IsRainHour	Trips
Austin	Commuter	0	169533
Austin	Commuter	1	8281
Austin	Leisure	0	1046341
Austin	Leisure	1	39673
Los Angeles	Commuter	0	235381
Los Angeles	Commuter	1	4696
Los Angeles	Leisure	0	846626
Los Angeles	Leisure	1	15365
Pittsburgh	Commuter	0	44601
Pittsburgh	Commuter	1	4845
Pittsburgh	Leisure	0	399426
Pittsburgh	Leisure	1	24049
Portland	Commuter	0	136400
Portland	Commuter	1	18214
Portland	Leisure	0	1027966
Portland	Leisure	1	102332

Chi-Square Test of Independence:

MetroArea	ChiSquare	df	p_value	CramersV	MinExpected
Austin	421.95805	1	9.15E-94	0.018272192	6746.877
Los Angeles	31.64227	1	1.85E-08	0.005358332	4370.134
Pittsburgh	1309.90894	1	7.94E-287	0.052629138	3020.997
Portland	1189.50859	1	1.16E-260	0.030426157	14505.351

SQL Code:

```
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips;

CREATE TEMPORARY TABLE ClassifiedTrips AS
SELECT
   TripID,
   MetroID,
   StartDate,
   StartTime,
   StartDayOfWeek,
   FromStationID,
   ToStationID,
   UserType,
   TripDuration_Minutes,
   CASE
      WHEN StartDayOfWeek BETWEEN 1 AND 5
          AND UserType = 'Subscriber'
          AND (
              (StartTime BETWEEN '06:30:00' AND '09:30:00')
              OR (StartTime BETWEEN '16:00:00' AND '19:00:00')
             )
          AND FromStationID <> ToStationID
      THEN 'Commuter'
      ELSE 'Leisure'
   END AS TripCategory
FROM Rentals;


--1) Hourly usage by metro & trip category
SELECT
   M.CoreCity AS MetroArea,
   C.TripCategory,
   HOUR(C.StartTime) AS StartHour,
   COUNT(*) AS TripCount,
   ROUND(
      100.0 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY M.CoreCity, C.TripCategory),
      2
   ) AS PercentWithinCategory
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.TripCategory, StartHour
ORDER BY M.CoreCity, C.TripCategory, StartHour;

--2) Daily usage by metro & trip category
```

```sql
SELECT
   M.CoreCity AS MetroArea,
   C.TripCategory,
   D.DayName_en AS DayOfWeek,
   COUNT(*) AS TripCount,
   ROUND(
      100.0 * COUNT(*) / SUM(COUNT(*)) OVER (PARTITION BY M.CoreCity, C.TripCategory),
      2
   ) AS PercentWithinCategory
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
JOIN DaysOfTheWeek D ON C.StartDayOfWeek = D.DayID
GROUP BY M.CoreCity, C.TripCategory, DayID, DayOfWeek
ORDER BY M.CoreCity, C.TripCategory, DayID;

--3) Monthly usage by metro & trip category
SELECT
   M.CoreCity AS MetroArea,
   C.TripCategory,
   YEAR(C.StartDate) AS YEAR,
   MONTH(C.StartDate) AS MONTH,
   COUNT(*) AS TripCount
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.TripCategory, YEAR, MONTH
ORDER BY M.CoreCity, C.TripCategory, YEAR, MONTH;

--4) Std. deviation across months (share of trips)
WITH MonthlyShares AS (
   SELECT
      M.CoreCity AS MetroArea,
      YEAR(C.StartDate) AS YEAR,
      MONTH(C.StartDate) AS MONTH,
      ROUND(
         100.0 * SUM(CASE WHEN C.TripCategory = 'Commuter' THEN 1 ELSE 0 END) /
COUNT(*),
         2
      ) AS CommuterSharePct
   FROM ClassifiedTrips C
   JOIN MetroArea M ON C.MetroID = M.MetroID
   GROUP BY M.CoreCity, YEAR, MONTH
)
SELECT
   MetroArea,
```

```
    STDDEV(CommuterSharePct) AS StdDevMonthlyCommuterSharePct
FROM MonthlyShares
GROUP BY MetroArea
ORDER BY MetroArea;


-- Rain Table
DROP TEMPORARY TABLE IF EXISTS TripsWithWeather;

CREATE TEMPORARY TABLE TripsWithWeather AS
SELECT
    T.MetroID,
    T.StartDate,
    T.StartHour,
    T.TripCategory,
    T.TripCount,
    W.Rain_mm_1h,
    CASE WHEN W.Rain_mm_1h > 0 THEN 1 ELSE 0 END AS IsRainHour
FROM TripsHourly T
JOIN Weather W
  ON T.MetroID = W.MetroID
 AND T.StartDate = W.DateOfReading
 AND T.StartHour = W.HourOfReading;

--5) Rain Resilience and Participation Ratio (Commuter vs Leisure, by Metro)
WITH TripRainShare AS (
    SELECT
        T.MetroID,
        T.TripCategory,
        100.0 * SUM(CASE WHEN T.IsRainHour = 1 THEN T.TripCount ELSE 0 END) /
SUM(T.TripCount) AS PctTripsInRain
    FROM TripsWithWeather T
    GROUP BY T.MetroID, T.TripCategory
),
RainHourShare AS (
    SELECT
        W.MetroID,
        100.0 * SUM(CASE WHEN W.Rain_mm_1h > 0 THEN 1 ELSE 0 END) / COUNT(*) AS
PctRainyHours
    FROM Weather W
    GROUP BY W.MetroID
)
SELECT
    M.CoreCity AS MetroArea,
```

```
    TR.TripCategory,
    ROUND(TR.PctTripsInRain, 2) AS PctTripsInRainHours,
    ROUND(RH.PctRainyHours, 2) AS PctRainyHours,
    ROUND(TR.PctTripsInRain / NULLIF(RH.PctRainyHours, 0), 3) AS RainParticipationRatio
FROM TripRainShare TR
JOIN RainHourShare RH ON TR.MetroID = RH.MetroID
JOIN MetroArea M ON TR.MetroID = M.MetroID
ORDER BY M.CoreCity, TR.TripCategory;


--6) Rain Contingency Table
SELECT
    M.CoreCity AS MetroArea,
    T.TripCategory,
    T.IsRainHour,
    SUM(T.TripCount) AS Trips
FROM TripsWithWeather T
JOIN MetroArea M ON T.MetroID = M.MetroID
GROUP BY M.CoreCity, T.TripCategory, T.IsRainHour
ORDER BY M.CoreCity, T.TripCategory, T.IsRainHour;
```